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CS23633 – Cloud Computing Laboratory**

Experiment 3: Setting Up Hybrid Cloud Connectivity Using Cisco Packet Tracer

AIM:

To simulate a secure hybrid cloud connection between an on-premises data center and a cloud network using routers, switches, subnets, and routing in Cisco Packet Tracer.

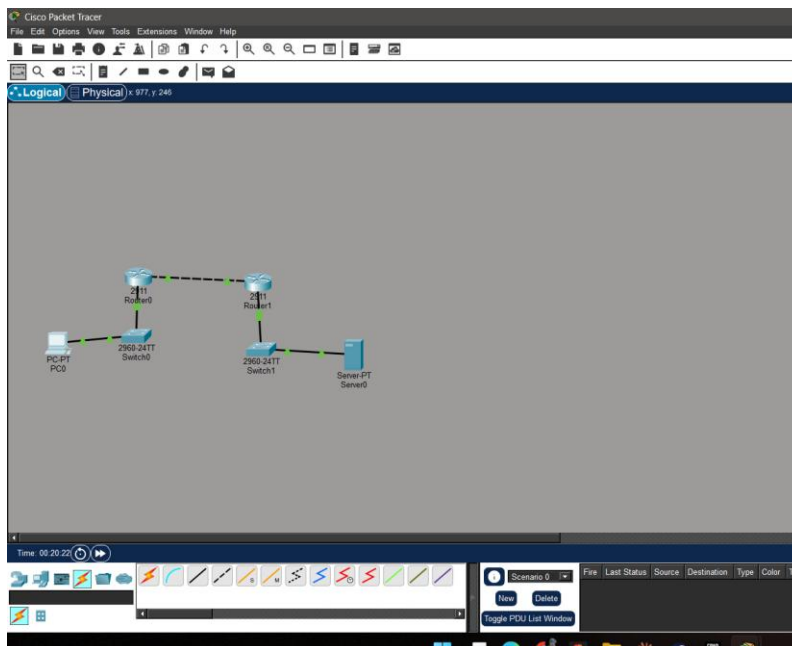
What is Hybrid Cloud?

A hybrid cloud is an IT architecture that combines on-premises infrastructure (private cloud) with third-party public cloud services, allowing data and apps to be shared between them

PROCEDURE:

Step 1: Create the Topology

- Add 2 Routers, 2 Switches, 1 PC, and 1 Server in Cisco Packet Tracer.
- Connect them using Copper Straight-Through cables.



Step 2: Configure Router0

- Configure GigabitEthernet0/0
 - IP Address: 192.168.1.1
 - Subnet Mask: 255.255.255.0
- Configure GigabitEthernet0/1

- IP Address: 10.10.10.1
- Subnet Mask: 255.255.255.252
- Enable both interfaces using no shutdown.

```

Router0
Physical Config CLI Attributes

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 10.10.10.1 YES manual up down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 10.10.10.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router(config)#

```

Step 3: Configure Router1

- Configure GigabitEthernet0/0
 - IP Address: 10.0.0.1
 - Subnet Mask: 255.255.255.0
- Configure GigabitEthernet0/1
 - IP Address: 10.10.10.2

- Subnet Mask: 255.255.255.252
- Enable both interfaces using no shutdown.

```

Router1
Physical Config CLI Attributes

DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable
% Incomplete command.
Router(config)#configure terminal
^
% Invalid input detected at '^' marker.
Router(config)#configure terminal
^
% Invalid input detected at '^' marker.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.0.1 YES manual up up
GigabitEthernet0/1 10.10.10.2 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router(config)#

```

Step 4: Verify Router Configuration

- Use: do show ip interface brief
- Ensure all required interfaces are in the up/up state.

Step 5: Configure PC0

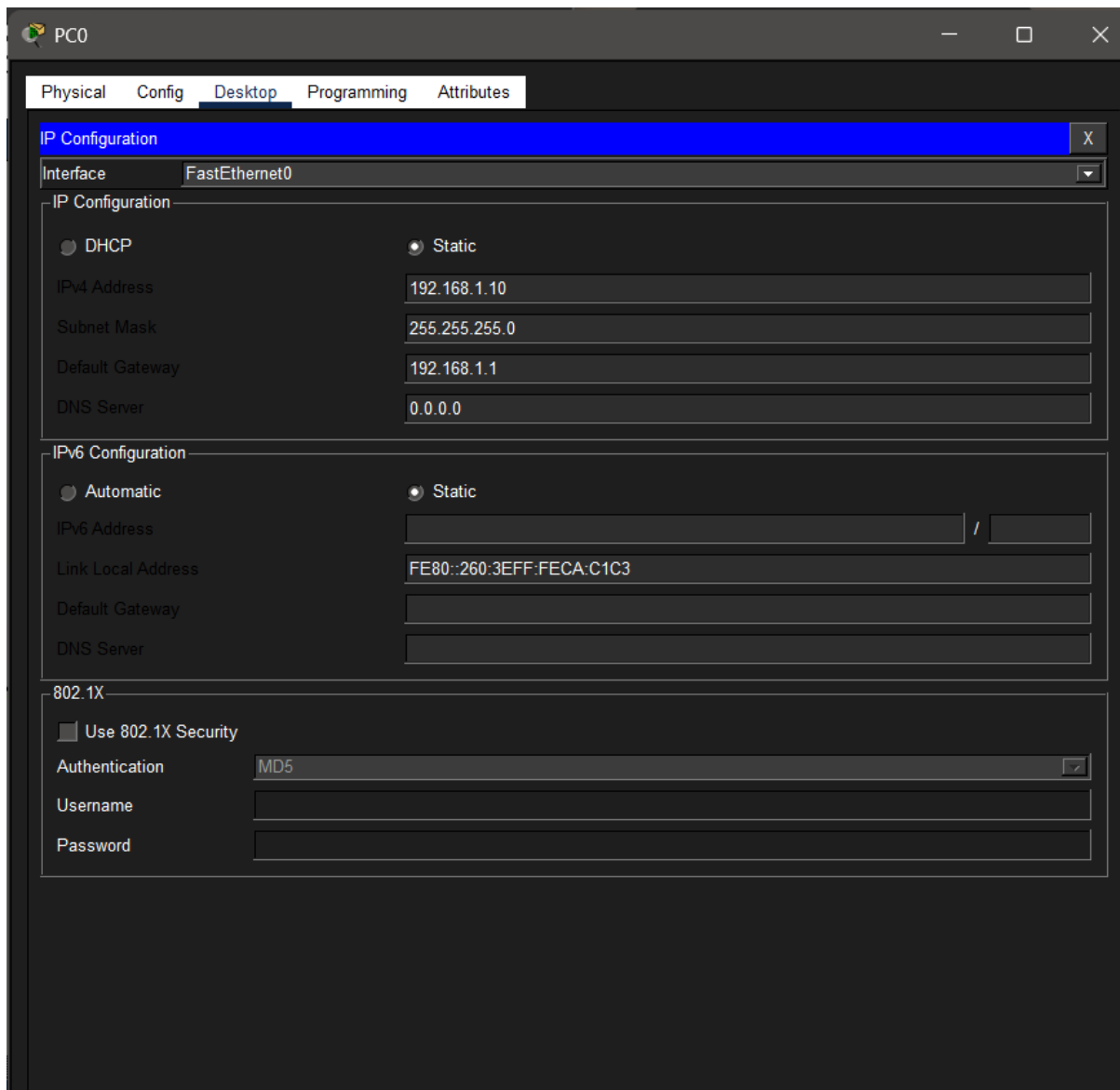
- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

The screenshot shows the 'Server0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted with a blue bar. Below it, the 'IP Configuration' panel shows 'Static' selected for the IP type. The IPv4 Address is set to 10.0.0.10, Subnet Mask to 255.255.255.0, and Default Gateway to 10.0.0.1. The DNS Server is set to 0.0.0.0. The 'IPv6 Configuration' panel shows 'Static' selected, with the IPv6 Address field empty and the Link Local Address set to FE80::260:2FFF:FE68:C7BD. The '802.1X' panel shows 'Use 802.1X Security' unchecked, with the Authentication dropdown set to MD5. The Username and Password fields are empty.

Section	Option	Value
IP Configuration	Static	
	IPv4 Address	10.0.0.10
	Subnet Mask	255.255.255.0
	Default Gateway	10.0.0.1
	DNS Server	0.0.0.0
IPv6 Configuration	Static	
	IPv6 Address	
	Link Local Address	FE80::260:2FFF:FE68:C7BD
	Default Gateway	
	DNS Server	
802.1X	Use 802.1X Security	Unchecked
	Authentication	MD5
	Username	
	Password	

Step 6: Configure Server0

- IP Address: 10.0.0.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.0.0.1



Step 7: Configure Static Routing

- On Router0: ip route 10.0.0.0 255.255.255.0 10.10.10.2
- On Router1: ip route 192.168.1.0 255.255.255.0 10.10.10.1

Step 8: Save the Configuration

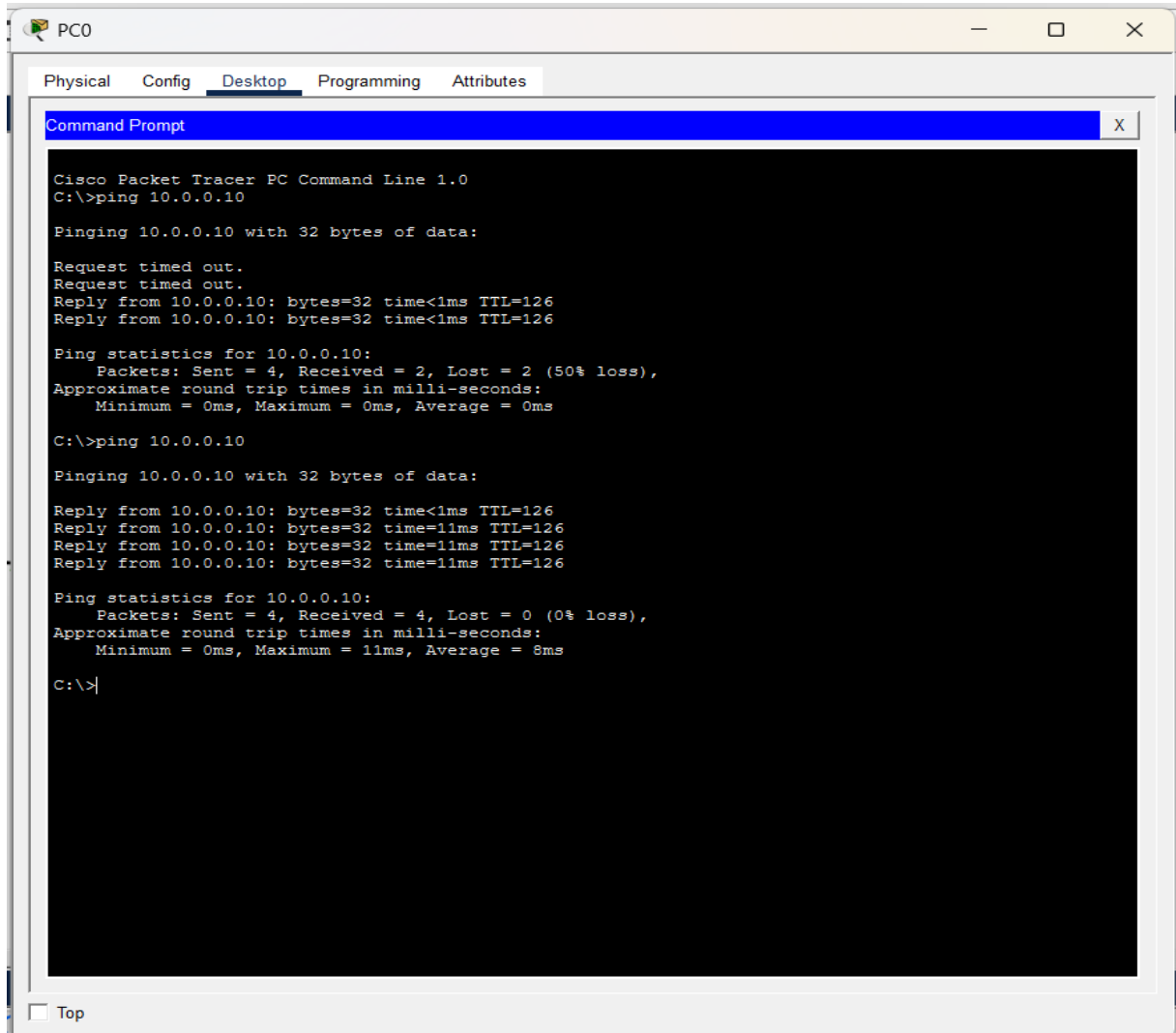
- Use: copy running-config startup-config
- Press Enter when prompted.

Step 9: Verify Routing Tables

- Use: show ip route
- Verify that the static routes (S) are displayed.

Step 10: Test Connectivity

- Open **PC0** → **Desktop** → **Command Prompt**.
- Execute: ping 10.0.0.10
- If the first attempt times out, ping once again.



```
PC0
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Reply from 10.0.0.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time<1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=11ms TTL=126
Reply from 10.0.0.10: bytes=32 time=11ms TTL=126
Reply from 10.0.0.10: bytes=32 time=11ms TTL=126

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 8ms

C:\>
```

Step 11: Save the Packet Tracer File

- Click **File** → **Save As**.
- Save the file with a suitable .pkt filename.

CONCLUSION:

Hybrid Cloud Connectivity was successfully established between the on-premises network (192.168.1.0/24) and the cloud network (10.0.0.0/24) using static routing in Cisco Packet Tracer.